

Comparison Between IPv4 and IPv6

Computer Networks – Student Study Material

Feature	IPv4	IPv6
Address length	Uses 32-bit addresses, providing about 4.3 billion addresses.	Uses 128-bit addresses, providing a vastly larger address space.
Address notation	Written in dotted-decimal form, for example, 192.168.1.1.	Written in hexadecimal form, for example, 2001:db8::1.
Header complexity	Header is variable, from 20 to 60 bytes, because of optional fields.	Header is fixed at 40 bytes, making packet processing simpler and faster.
Header checksum	Includes a header checksum for error detection.	Does not include a header checksum; error checking is handled by other layers.
Security	IPsec support is available but generally optional.	Provides standardized and built-in support for IPsec security features.
Autoconfiguration	Usually requires manual configuration or a DHCP server.	Supports Stateless Address Autoconfiguration (SLAAC), allowing devices to configure addresses automatically using router advertisements.
Address resolution	Uses ARP to map IP addresses to MAC addresses.	Uses Neighbor Discovery Protocol (NDP) through ICMPv6 instead of ARP.
Broadcasting	Supports broadcast communication.	Does not use broadcast; it uses multicast and anycast instead.
Fragmentation	Routers and source devices can fragment packets.	Only the source device can fragment packets; routers do not perform fragmentation.
NAT requirement	NAT is commonly used because IPv4 addresses are limited.	NAT is generally unnecessary because of the large address space.